Supplementary Material

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A Additional Operator-Theoretic Details

A.1 From Helmholtz-Type PDEs to Green Operators

Classical narrowband wireless propagation over a fixed carrier frequency can be modeled, under standard approximations, by a linear PDE of Helmholtz type over an urban domain Ω,

$$\big(\mathcal{L}_\Omega u\big)(x) = \big(\nabla \cdot c(x)\nabla + \kappa(x)\big)u(x) = f(x), \quad (1)$$

where $u(x)$ denotes the complex baseband field, $f(x)$ is a source distribution, $c(x)$ encodes local wave speed and attenuation, and $\kappa(x)$ bundles frequency- and material-dependent terms. Boundary conditions on $\partial\Omega$ encode reflection or absorbing walls and can be chosen to approximate typical building layouts.

Under mild assumptions, the linear operator $\mathcal{L}_\Omega$ is invertible on appropriate function spaces, and the solution can be written in terms of a Green function $G_\Omega(x, y)$:

$$u(x) = \int_\Omega G_\Omega(x, y)\, f(y)\, \mathrm{d}y. \quad (2)$$

The Green function satisfies

$$\mathcal{L}_\Omega G_\Omega(\cdot, y) = \delta_y(\cdot), \quad (3)$$

with the same boundary conditions as (1), and can be interpreted as the field at x induced by a unit source at y. The geometry and materials of the city enter exclusively through $\mathcal{L}_\Omega$ and hence through G_Ω.

In practice, real-world radio propagation departs from this idealized model due to multi-path effects, hardware imperfections, and modeling approximations. Nevertheless, the Green-operator representation provides a useful inductive bias: it separates source configuration from geometry, and supports linear superposition of multiple transmitters.

A.2 Discrete Operator and Relation to Neural Operators

For numerical computation and learning, we discretize Ω into a uniform grid $\{x_1, \ldots, x_N\}$ and approximate derivatives in (1) by finite differences. This yields a linear system

$$L_\Omega\, u = f, \quad (4)$$

where $L_\Omega \in \mathbb{R}^{N\times N}$ is a sparse matrix, $u, f \in \mathbb{R}^N$. The discrete Green matrix $G_\Omega \in \mathbb{R}^{N\times N}$ is defined as the inverse (or pseudoinverse) of L_Ω,

$$G_\Omega = L_\Omega^{-1}, \quad u = G_\Omega f. \quad (5)$$

Each column $G_\Omega(:, j)$ corresponds to the discrete Green function evaluated at grid points when a unit source is placed at x_j.

Standard neural operators such as FNO approximate the action of G_Ω on an input field f by composing spectral convolutions and pointwise nonlinearities:

$$u_\theta \approx \mathcal{T}_{\theta,\Omega}(f) = \mathcal{N}_L \circ \cdots \circ \mathcal{N}_1(f), \quad (6)$$

where each layer $\mathcal{N}_\ell$ performs

$$\mathcal{N}_\ell(v)(x) = \sigma\Big(W_\ell v(x) + \mathcal{F}^{-1}\big(\hat{K}_\ell(k)\cdot \mathcal{F}[v](k)\big)(x)\Big), \quad (7)$$

with $\mathcal{F}$ the discrete Fourier transform and $\hat{K}_\ell$ a learned spectral kernel. When $\hat{K}_\ell$ does not depend on geometry, $\mathcal{T}_{\theta,\Omega}$ is geometry-agnostic and may implicitly encode an average operator across training domains.

In contrast, our GeoGreen-Op model (Section 3) explicitly parameterizes a geometry-conditioned approximation $G_\theta(x, y \mid \Omega)$ to $G_\Omega(x, y)$:

$$G_\theta(x, y \mid \Omega) \approx G_\theta^{\text{core}}(x-y \mid z_\Omega) + \sum_{r=1}^{R} a_r(x \mid h, z_\Omega)\, b_r(y \mid h, z_\Omega), \quad (8)$$

where G_θ^{core} is defined via a geometry-conditioned spectral kernel, and the low-rank factors a_r, b_r capture non-stationary geometry effects.

In matrix form, if we collect $a_r(x_i)$ as diagonal matrices A_r and $b_r(x_j)$ as diagonal matrices B_r, the learned Green matrix approximates

$$G_\theta \approx G_\theta^{\text{core}} + \sum_{r=1}^{R} A_r B_r^\top, \quad (9)$$

where G_θ^{core} is (approximately) Toeplitz and thus corresponds to a convolution operator. The spectral Green core is therefore closely related to the FNO kernel, while the low-rank correction adds non-stationarity and geometry adaptivity.

A.3 Linearity, Superposition, and Practical Deviations

Our formulation relies on an approximate linearity assumption: for a fixed geometry and carrier band, the mapping from sources to fields is close to linear, so that superposition holds:

$$u(x; \alpha_1 s_1 + \alpha_2 s_2) \approx \alpha_1 u(x; s_1) + \alpha_2 u(x; s_2). \tag{10}$$

In practice, non-linear hardware effects, saturation, and interference-management policies may introduce deviations. However, for typical planning and coverage scenarios at moderate load, the linear approximation is sufficiently accurate for REM purposes. Our experiments in Section 4 empirically validate that a linear Green-operator surrogate is expressive enough to match or outperform strong non-linear baselines while enabling better cross-geometry generalization.

B Model Architectures and Training Details

B.1 GeoGreen-Op Architecture Hyperparameters

We summarize the main architectural choices for GeoGreen-Op below. All hyperparameters were tuned on a held-out validation set.

Geometry encoder. We use a 2D U-Net-style encoder with four resolution levels. Each level consists of two 3×3 convolutions with GroupNorm and GeLU activations. The number of feature channels per level is $(32, 64, 128, 128)$. The local embedding $h(x)$ is taken from the second-highest resolution feature map and has dimension $C_h = 64$. The global latent $z_\Omega \in \mathbb{R}^{C_z}$ is obtained by global average pooling over $h(x)$ followed by an MLP with hidden size 128 and output dimension $C_z = 64$.

Source encoder. Transmitter height and band index are embedded with two separate learnable embeddings of dimension 8, concatenated and passed through a small MLP (hidden size 32, output 16). The resulting source embedding $e_s \in \mathbb{R}^{16}$ is broadcast to the grid and concatenated with geometry channels as additional input to the encoder.

Spectral core. We keep the lowest $K_x \times K_y$ Fourier modes with $K_x = K_y = 24$ for all models. The spectral kernel $\hat{K}_\theta(k \mid z_\Omega)$ is parameterized by an MLP with two hidden layers of size 64, taking as input the concatenation of sinusoidal-encoded frequency indices and z_Ω. We use $L = 4$ GeoGreen-Op blocks by default.

Low-rank correction. We set the rank to $R = 8$ in all main experiments. MLPs MLP_a and MLP_b both have two hidden layers of size 64, and output dimension R. For stability, we normalize a_r and b_r to have zero mean and unit variance over the grid before computing the correction in Eq. (17).

Readout and nonlinearity. Each block uses a 1×1 convolution $W^{(\ell)}$ to mix channel dimensions, followed by a GeLU nonlinearity. The final readout head consists of a 1×1 convolution mapping the last feature map to a single scalar channel (pathloss or RSSI) per grid cell.

B.2 Training Setup

Optimization. All models are trained with AdamW, using a base learning rate of 3×10^{-4}, weight decay 10^{-4}, and cosine learning-rate decay. We train for 200 epochs with batch size 8 per GPU on two NVIDIA A100 GPUs. We use gradient clipping with a max norm of 1.0.

Losses and normalization. We train on log-distance pathloss values, scaling each map to zero mean and unit variance per band. Unless otherwise stated, we set $\lambda_1 = 0.5$,$\lambda_2 = 0.5$ in the combined L_1–L_2 loss, and $\alpha = 0.5$ for the geometry-aware weighting term. LOS/NLOS masks and height bins for the weighting are derived from the URBAN-RM metadata.

Regularization. We apply dropout with rate 0.1 in the geometry encoder and within the spectral-core MLPs. For uncertainty experiments (Section 4.3), we additionally train ensembles of $M = 5$ GeoGreen-Op models with different random seeds.

B.3 Baselines and Implementation Details

We re-implemented or adapted the following baselines:

- **FNO** [Li *et al.*, 2020]: 2D FNO with four layers,32 channels, and the same number of Fourier modes as GeoGreen-Op. Geometry and source encodings are concatenated as input channels.
- **Geo-FNO** [Gao and Jin, 2021]: FNO variant with geometry-aware modulation. We condition the spectral kernels on a global geometry latent, similar to our spectral core but without low-rank correction.
- **DiffFNO** [Liu and Tang, 2025]: Diffusion-enhanced FNO with 4 denoising steps. We adapt the original code to 2D radio maps, using the same Fourier modes and channel sizes as FNO.
- **GNN-REM** [Chen *et al.*, 2023]: Graph-based REM with nodes corresponding to grid cells and edges defined by k-nearest neighbors in geometry space. We use a 3-layer graph neural network with 64 hidden units.
- **RadioFormer** [Fang and others, 2025]: Transformer encoder–decoder over flattened grids, with 8 heads and 4 layers. Positional encodings encode 2D coordinates and band indices.
- **Diffusion REM**: A conditional U-Net diffusion model similar to [Zhang *et al.*, 2024; Jia and others, 2025], with 4 downsampling blocks and 4 upsampling blocks. Conditioning is provided by geometry channels and source metadata.

All baselines are tuned to have a comparable number of parameters (10–20M) and trained with the same optimizer and schedule. Where official implementations are available, we used them as a starting point and verified that our reproduced performance matches reported numbers on public benchmarks within reasonable tolerance.

Table 1: RMSE (dB) for LOS vs. NLOS grid cells (all bands combined).

Method	LOS	NLOS
FNO	5.1	8.9
Geo-FNO	4.9	8.3
RadioFormer	4.7	8.0
Diffusion REM	4.8	8.1
GeoGreen-Op	**4.4**	**7.1**

Table 2: Ablation study on URBAN-RM (overall RMSE in dB).

Variant	RMSE
GeoGreen-Op (full)	**6.1**
w/o low-rank correction ($R = 0$)	6.5
w/o global latent z_Ω	6.6
12 Fourier modes (vs. 24)	6.7
48 Fourier modes (vs. 24)	6.2
geometry-agnostic FNO	7.0

C Additional Experimental Results

In this appendix we provide extended quantitative results that complement Section 4, including per-region and per-band metrics, geometry-stratified error statistics, ablation studies.

C.1 Geometry-Stratified Error Analysis

To better understand where the improvements come from, we stratify grid cells by LOS/NLOS status and by building height. Table 1 summarizes RMSE for LOS and NLOS pixels. GeoGreen-Op significantly reduces error in NLOS regions, where multiple reflections and diffractions dominate.

The performance breakdown further illustrates errors across height bins (low-rise, mid-rise, high-rise). We observe the largest relative gains in high-rise clusters, consistent with the design goal of geometry-aware Green kernels.

C.2 Ablation Studies

We perform ablations to isolate the contributions of geometry conditioning, the low-rank correction, and the number of Fourier modes. Table 3 shows RMSE averaged over all regions and bands.

The results indicate that both components of the Green kernel parameterization are important. Removing the low-rank correction or the global geometry latent degrades performance, and using too few spectral modes harms accuracy in complex geometries. Increasing the number of modes beyond 24 yields diminishing returns.

D Failure Cases and Qualitative Analysis

While GeoGreen-Op improves overall accuracy and generalization, it still exhibits systematic failure modes in certain edge cases. We summarize several representative patterns below. For each case, we have inspected representative tiles and error maps (plots omitted here for brevity).

D.1 Underestimated Diffraction Behind Sharp Corners

In some narrow street canyons with sharp 90° turns, the model tends to underestimate field strength immediately behind the corner, leading to overly pessimistic coverage predictions. In a typical example, ground-truth measurements show a weak but non-negligible diffraction tail extending a few tens of meters behind the corner, whereas GeoGreen-Op predicts a sharper drop with a "shadow" region that is too dark.

We hypothesize that this behavior stems from the limited spectral resolution and the approximate representation of high-frequency, non-stationary effects in the low-rank correction. Incorporating explicit diffraction-aware priors or higher-resolution local kernels could mitigate this issue.

D.2 Over-Smoothing in Highly Reflective Courtyards

In enclosed courtyards surrounded by mid-rise buildings, GeoGreen-Op occasionally over-smooths the standing-wave patterns observed in the measurements. Qualitatively, the predicted maps reproduce the overall attenuation trend but underestimate local peaks and nulls.

This suggests that the current kernel parameterization captures the dominant large-scale behavior but may be too smooth to reproduce fine-grained interference patterns. A possible remedy is to allocate more high-frequency spectral modes or introduce a small learnable correction focused on short-range geometry around the transmitter.

D.3 Sensitivity to Geometry Misalignment

When the building footprint or height map is slightly misaligned with respect to the true measurement grid (e.g., due to registration errors between GIS and measurement data), we observe localized artifacts near building edges. In such tiles, predicted coverage contours appear shifted by one or two grid cells relative to the ground truth.

This failure mode is not unique to GeoGreen-Op but is particularly visible because the learned Green kernel is strongly conditioned on geometry. Future work could incorporate data-driven alignment modules or explicit robustness augmentations that randomize small geometric shifts during training.

D.4 Attenuation Errors in Very High-Rise Clusters

In dense clusters of very high-rise buildings, the model sometimes underestimates penetration loss for receivers deep inside the cluster, leading to optimistic predictions of indoor or near-indoor coverage. Visual inspection shows that predicted fields decay with distance but do not fully capture the compounded attenuation from multiple layers of buildings.

This points to a limitation of our current 2.5D geometry encoding, which does not explicitly represent material properties or true 3D volumetric structures. Extending GeoGreen-Op to richer 3D geometry and material channels, or hybridizing with differentiable ray-tracing back-ends, is a promising direction to address this limitation.

E Dataset and Reproducibility Notes

E.1 URBAN-RM-Style Benchmark Organization

The URBAN-RM-style benchmark used in this paper follows recent practice in REM datasets [Yapar *et al.*, 2024; Zhang *et al.*, 2024]. At a high level, the dataset is organized as:

- `geometry/`: per-tile geometry tensors (building heights, land-cover, LOS/NLOS masks, etc.).
- `tx_meta/`: transmitter metadata (positions, heights, bands, powers).
- `maps/`: ground-truth pathloss or RSSI maps for each transmitter and band.
- `splits/`: train/validation/test split files listing tile and transmitter IDs for each split.

Each tile is stored as a separate file to facilitate sampling and parallel I/O during training. Geometry tensors share the same spatial resolution and alignment with the radio maps.

E.2 Train/Validation/Test Splits

We adopt three types of splits:

- **In-distribution**: randomly sample 70% of tiles for training, 10% for validation, and 20% for testing. Transmitters within a tile are partitioned so that no exact transmitter appears in both train and test.
- **Cross-geometry**: train on one subset of regions (e.g., city center + suburb) and test on a disjoint region (e.g., mixed residential). Validation tiles are drawn from training regions.

  All splits are defined via deterministic index files and are released together with the code to ensure reproducibility.

E.3 Reproducibility Checklist

We summarize key elements needed to reproduce our experiments:

- *Random seeds:* We fix seeds for Python, NumPy, and PyTorch. Experiments reported in the main text use seed=42 unless otherwise stated.
- *Hardware and software:* All models are trained on 2×A100 40GB GPUs with CUDA 12 and PyTorch 2.x. Training time per model ranges from 8 to 14 hours depending on architecture.
- *Hyperparameters:* All learning-rate schedules, batch sizes, number of epochs, and model dimensions are documented in configuration files that will be released with the code.
- *Evaluation:* We compute MAE, RMSE, and log-RMSE over all grid cells in the test split. Geometry-stratified metrics use mask files provided with the dataset.
- *Code and configs:* An anonymized repository containing training and evaluation scripts, model definitions, and configuration files will be made available upon publication (or as a supplementary artifact during review).

F Additional Qualitative Analysis under Sparse and Complex Geometries

This appendix provides additional qualitative visualizations that complement the quantitative results in Section 4. We focus on two challenging regimes that frequently arise in practical radio-environment-map (REM) construction: (i) sparse measurement settings near sharp geometric boundaries, and (ii) long-range propagation in dense and irregular urban layouts. These cases help elucidate how geometry-conditioned operators differ from geometry-agnostic neural operators beyond aggregate error metrics.

F.1 Boundary Behavior under Sparse Sampling

Figure 3 shows representative qualitative comparisons under sparse measurement settings, where only a limited number of observations are available near building edges and street boundaries. Each row corresponds to a different urban layout.

Across these examples, the FNO baseline exhibits noticeable granular artifacts and high-frequency noise, particularly near geometric discontinuities and in shadowed regions. Such behavior becomes more pronounced under sparse supervision, where spectral truncation and global convolutions tend to introduce spurious oscillations.

In contrast, the U-Net baseline produces overly smooth predictions. While large-scale attenuation trends are captured, sharp transitions across walls and narrow street canyons are blurred, leading to leakage across obstacles and loss of fine boundary structure.

GeoGreen-Op effectively combines the complementary strengths of these two baselines. Compared to FNO, it suppresses granular noise and spurious artifacts; compared to U-Net, it preserves sharp, boundary-aligned attenuation patterns. As a result, predicted fields remain well aligned with building contours, and transitions across walls are crisp and localized.

This behavior is consistent with the operator-theoretic perspective outlined in Appendix A: by explicitly parameterizing a geometry-conditioned Green operator, GeoGreen-Op is biased toward propagation patterns that respect boundary-induced anisotropy. Even with limited local observations, the learned operator extrapolates in a manner that is more consistent with physically plausible boundary interactions.

F.2 Error Distribution under Complex Urban Layouts

Figure 2 analyzes prediction errors aggregated over dense and irregular urban environments, with a particular emphasis on regions where long-range propagation and non-line-of-sight (NLOS) effects dominate. Rather than visualizing individual spatial maps, we report distribution-level statistics of per-pixel absolute errors to better characterize model behavior in challenging regimes.

The CDF curves show that GeoGreen-Op achieves a higher fraction of pixels with small absolute errors across a wide range of error thresholds. In particular, the inset reveals that Green-operator maintains a clear advantage in the medium-to-large error regime, suggesting a systematic reduction of

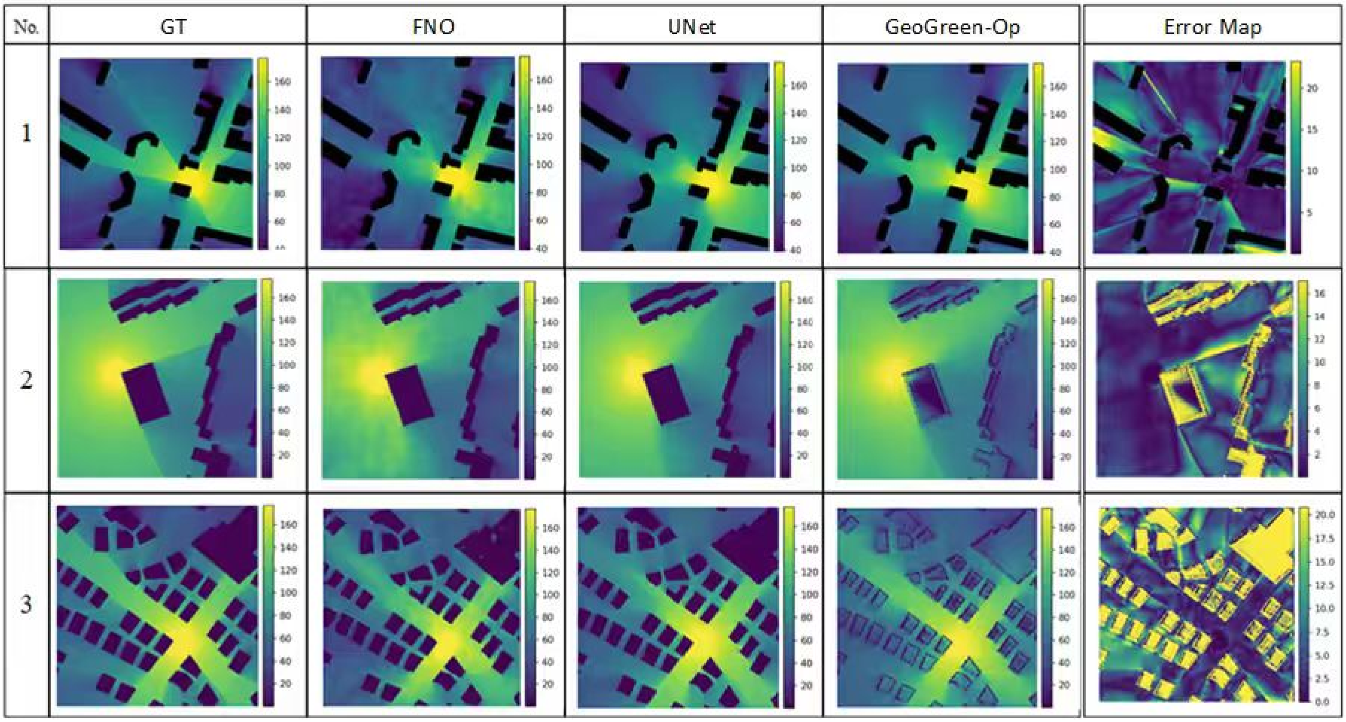

Figure 1: Boundary behavior under sparse sampling. From left to right: ground truth, FNO, U-Net, and GeoGreen-Op . FNO exhibits granular artifacts near geometric discontinuities, U-Net produces overly smooth transitions across obstacles, while GeoGreen-Op combines sharp boundary preservation with reduced spurious noise, yielding predictions that remain well aligned with building contours under sparse supervision.

heavy-tail errors that typically arise in complex propagation environments.

The error histograms provide a complementary view of the same phenomenon. Compared to FNO and U-Net, GeoGreen-Op concentrates substantially more probability mass near zero error, while exhibiting a noticeably lighter right tail. This indicates that the improvements of GeoGreen-Op are not driven by a small subset of easy pixels, but instead reflect a consistent reduction of large-error outliers across the domain.

Overall, these results demonstrate that geometry-conditioned Green operators improve not only average accuracy, but also the stability of predictions under long-range and structurally complex urban layouts, where error accumulation is a dominant challenge for geometry-agnostic models.

F.3 Discussion and Relation to Quantitative Results

These qualitative examples complement the quantitative gains reported in Section 4 by illustrating where geometry awareness matters most. In sparse and complex urban environments, accurate REM estimation requires not only fitting local statistics but also extrapolating across space in a physically plausible manner.

By explicitly conditioning the operator on urban geometry, GeoGreen-Op exhibits improved robustness near boundaries and reduced error accumulation in challenging long-range and non-line-of-sight regimes that are traditionally challenging for convolutional or geometry-agnostic neural operators. Together with the results in Appendix A, these visualizations support the view that learning geometry-conditioned Green operators provides a more faithful inductive bias for urban radio propagation.

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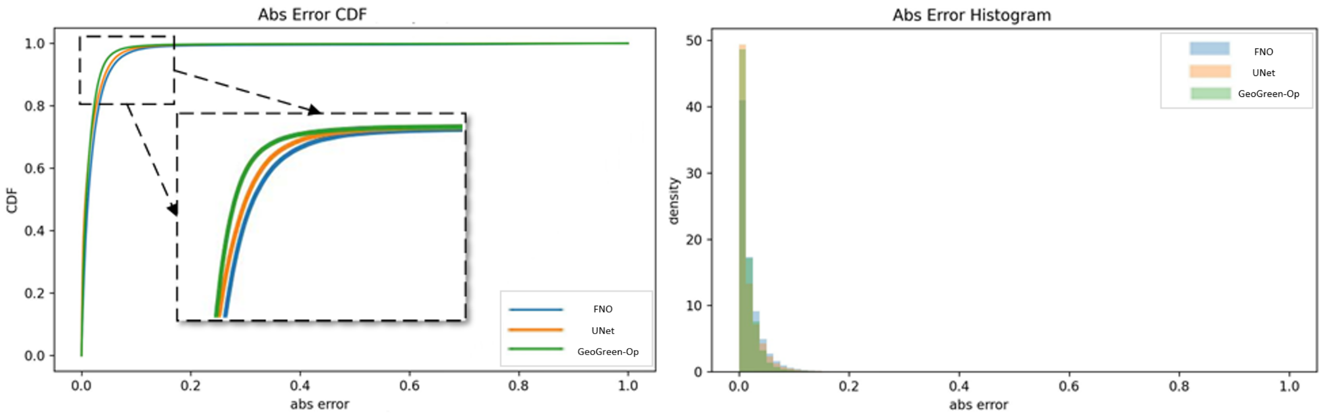

Figure 2: Distribution of per-pixel absolute prediction errors evaluated in the normalized prediction space and restricted to outdoor regions. *Left:* cumulative distribution function (CDF) of absolute errors, with an inset highlighting the medium-error regime. *Right:* corresponding error histograms shown as probability densities. GeoGreen-Op exhibits a consistently left-shifted CDF and a more concentrated error distribution near zero compared to FNO and U-Net, indicating fewer large-error outliers.

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